

Manthan Vijay Barhate

Portfolio: <https://barhatemanthan.github.io/portfolio/>
Github: <http://www.github.com/BarhateManthan>

Email: Manthanbarhate70@gmail.com
Mobile: +91 7385628262
LinkedIn: <http://www.linkedin.com/in/manthanbarhate>

EDUCATION

- Pimpri Chinchwad College of Engineering** Pune, India
B.Tech in Computer Science (AI-ML), CGPA: 8.58/10.00 Nov 2022 - Jul 2026
- P.O. Nahata College of Science** Bhusawal, India
Class XII (HSC), Percentage: 87.00% 2020 - 2022

SKILLS

- Languages:** Python, C++, JavaScript, SQL
- Frameworks:** NLTK, TensorFlow, Deep Learning, Node.JS (Express)
- Tools:** GIT, MySQL, MongoDB
- Platforms:** Linux, Windows, Azure

EXPERIENCE

- AAAI (Association for the Advancement of Artificial Intelligence)** Coordinator
Managed and volunteered the Agentic AI topics covering RAG and scalability of agents Jan 2025 - present
 - Organized AI tech talk series featuring Wipro's Director in AI discussing advanced agent architectures
- TE Connectivity** Freelancer
Developed ML algorithm to identify high-potential market qualified leads (MQLs) Jan 2025 - present
 - Analyzed customer activity data like cart additions, product views, and interaction timelines with TE
 - Created algorithms to identify top-k MQLs based on customer conversion patterns

PROJECTS

- HDFC Bank Annual Report RAG-Bot (Jan 2025):**
 - Tools:** LangChain, Llama 3.1, FAISS, Slack API, OCR
 - Designed a RAG-based chatbot into the Slack platform, attaining a 80% accuracy rate in providing context-relevant answers to user queries.
 - Established a custom OCR pipeline and tokenization techniques, enhancing embedding quality by 30%.
 - Architected a hybrid search system using FAISS vector database, minimizing search latency by 40%.
- EEG Brain Activity Classification (Nov 2024):**
 - Tools:** Python, TensorFlow, EfficientNet, ICA, Wavelet Transforms
 - Accomplished a KL divergence of 0.28 in classifying harmful brain activities into 5 seizure categories using EfficientNet.
 - Converted raw EEG signals into spectrograms using db2/CWT transforms, optimizing signal clarity by 25%.
 - Orchestrated ICA preprocessing, extracting relevant brain signals with 15% higher accuracy.
- Gen AI Sales Assistant (Aug 2024):**
 - Tools:** AWS S3, Node.js, Vultr API, MongoDB
 - Constructed a platform generating curated sales pitches, elevating customer engagement by 20%.
 - Utilized ML algorithms to analyze customer interactions, delivering actionable insights with 60% accuracy.
 - Spearheaded a backend system supporting 100+ users, including login, S3 buckets, and Gen-AI calls.
- SevaMitra - Streamlining Medical Documentation (Feb 2025):**
 - Tools:** Python, RAG, FAISS, OCR, Gemini, Langchain
 - Built a pipeline that synced dynamically from G-Drive to fetch latest documents to append the current vector store.
 - Built a chatbot backed with the vector database to give relevant answer to doctor's queries.
 - A real-time note addition feature to help doctors for patient's next clinical trial summarized by Gemini.

ACHIEVEMENTS

- Bagged **1st rank** in state-level PICT DATAQUEST datathon.
- Achieved **4th rank** in Cavista Hackathon.
- Achieved **10th rank** in the national hackathon BITS – INDOML.
- Secured **40th rank** in the Vultr Cloud National hackathon among 4500 participants.
- Scored **97.47th** percentile in MHTCET.

CERTIFICATIONS

- Andrew NG Deep Learning Specialization Nov 2024 – Ongoing 
- Oracle Gen-AI Professional Certificate Jul 2024 
- Andrew NG Machine Learning May – Jul 2024 
- AWS Cloud and AI ML Foundations Mar 2024 